

NORDIC
INTEGRATION
SUMMIT

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Tunneling into Azure Functions
Test Locally, Think Globally

By

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#nis25



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CONCEPTS



The Developer’s Dilemma: Testing in Isolation

Local development environment
- Safe, fast, reliable
- But isolated behind firewalls

External services want to call your app
- Power Automate needs a webhook
- GitHub sends push events
- Stripe posts payment callbacks
- Azure services expect HTTPS endpoints

Traditional testing cycle
- Slow
- Error-prone.
- Consumes time and resources.



When your code works locally… but the world can’t see it

From local host to the world - securely



Develop locally, test

What are tunnels

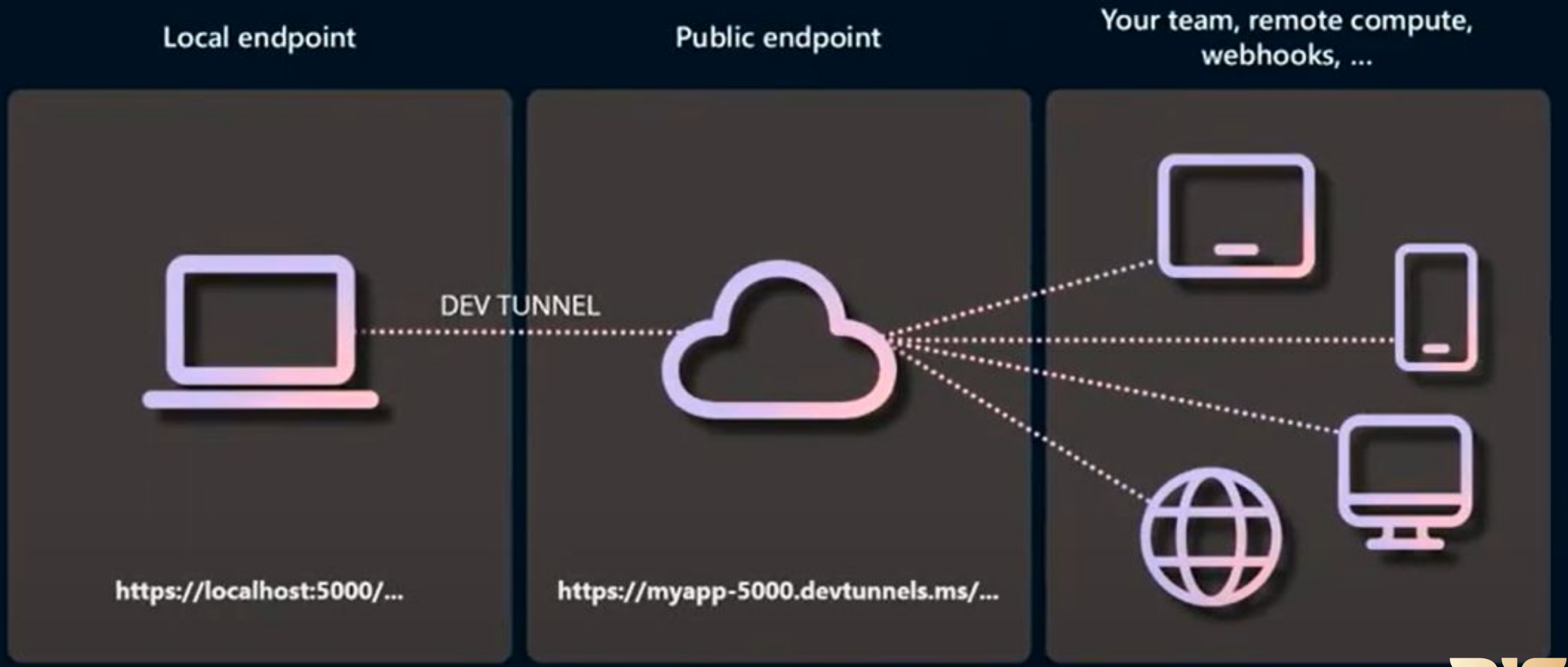
- A secure, temporary **public endpoint** for your local web app
- Works with **any local port or framework** (Functions, APIs, SPA, etc)
- No need for deployment — just connect and share
- Built into **Visual Studio** and the **Azure Dev CLI**
- Ideal for testing **webhooks, integrations, and cross-device access**

Expose by Beyond Code

CLOUDFLARE

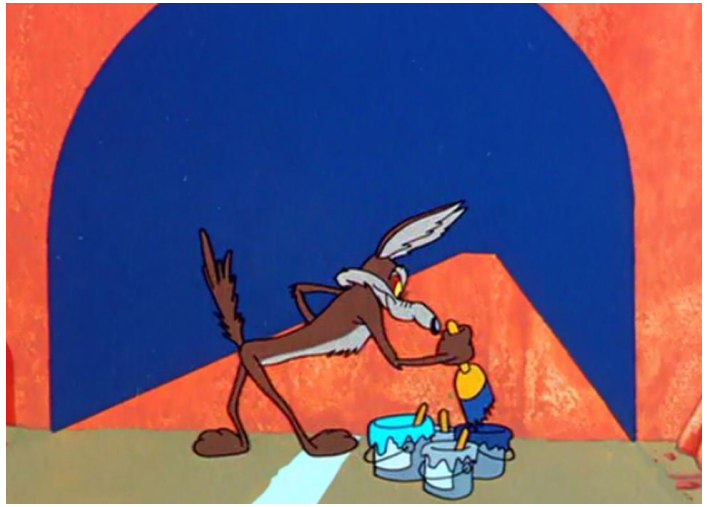
ngrok

How do tunnels work



How do tunnels work

run
`devtunnel host -p 7053`



create a public
HTTPS endpoint



traffic is securely forwarded
to the defined local port



Why use tunnels



Test webhooks and external services locally



Integrate Power Platform / Logic Apps / GitHub with local APIs



Secure, auditable, identity-based access



Share local apps for review or debugging



Skip deployment cycles during rapid dev

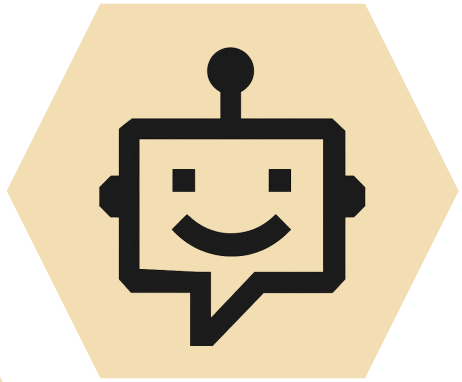
Where to use tunnels



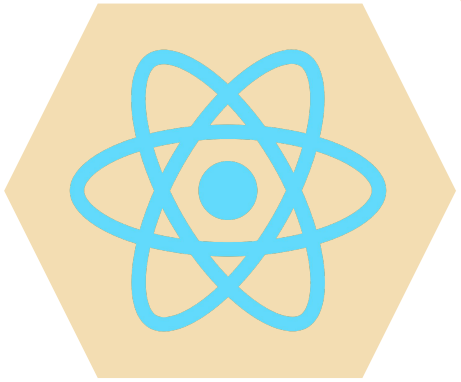
test triggers & webhooks locally



ASP.NET APIs for connections



chatbots or webhooks Teams, Github



front-end frameworks share live builds



remote validation in CI/CD

Security & access



- All tunnels use TLS (HTTPS)
- Identity: authenticated via Entra ID
- Access levels:
  Private Only you
  Org Signed-in org users
  Public Anyone with the link
- URLs are temporary & revocable

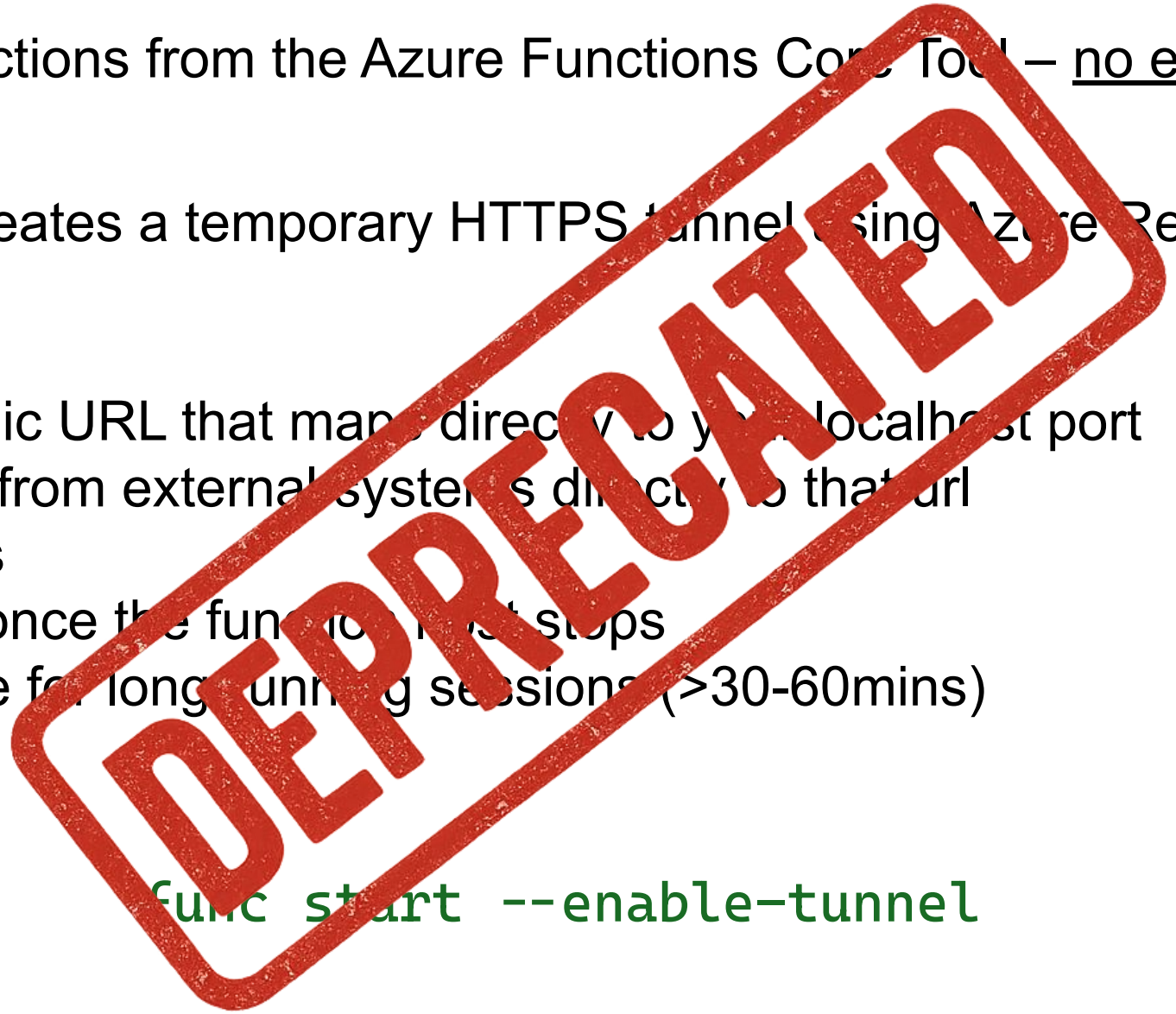

TOOLS

Azure Functions Tunnel

Testing azure functions from the Azure Functions Core Tool – no extra installs

Core Tools CLI creates a temporary HTTPS tunnel using Azure Relay under the hood

- Generates public URL that maps directly to your localhost port
- Send requests from external systems directly to that url
- Automatic https
- Tunnel closes once the function host stops
- Limit: not stable for long running sessions (>30-60mins)




```
func start --enable-tunnel
```

Azure Dev Tunnel

Best for developers using visual studio 2022+ or Visual Code

- Dev Tunnels create a secure, authenticated tunnel directly from your IDE using Azure identity
- Public URL generated under *.devtunnels.ms
- It can forward HTTP, websocket and other protocols
- Integrates with Azure AD (optional)

Security & access

Tips for authentication and access control



Login with the CLI



Grant access to your org



Configure auth redirection



Use a client access token



Pre-auth a client connection

ngrok

Classic, quick setup

- Ngrok creates a reverse proxy from your local port to an HTTPS url on their cloud relay
- Offers dashboards, request replay, authentication and traffic inspection
- Not the first, but the most famous
- No account/no login required, for authenticated mode uses ngrok token

CLOUDFLARE

Secure, outbound-only, zero-trust

- Lightweight agent runs locally and creates an outbound-only https connection to cloudflare
- Optional policies to protect the endpoint: SSO, JWT, device posture
- Authentication via cloudflare login available

Compare

Tool	Type	Setup Simplicity	Auth	Typical use
Azure Dev Tunnel	Microsoft	★★★★	Azure EntraID	IDE integration
ngrok	3rd party	★★★★★	token	Most popular
CloudFlare	3rd party	★★★	Cloudflare login	Secure + DNS
Expose.dev	Open-source	★★★	optional	Encrypted
LocalXpose	3rd party	★★★	token	Regional testing


DEMO

Azure Dev Tunnel

PREREQUISITES

```
winget install Microsoft.AzureCLI
winget install Microsoft.devtunnel
az login
devtunnel user login
```

STEP BY STEP

Func start
- http url – get the port

```
devtunnel create nis25demo --allow-anonymous
devtunnel host -p {port}
```
- result

```
    Hosting tunnel 'tunnel-1234' on port 7071
    Public URL:
https://green-field-1234.devtunnels.ms
    Inspection URL:
https://devtunnels.ms/inspect/1234
```

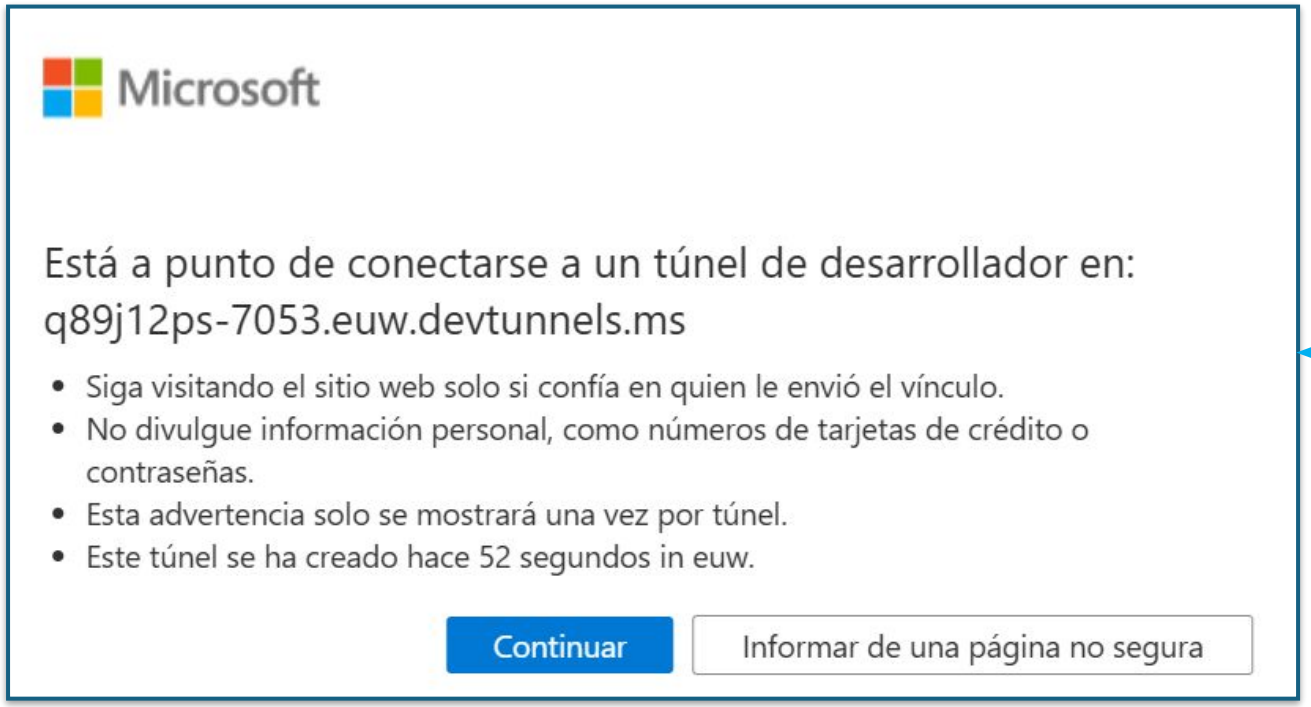
Use the public url provided – pautomate, postman…



Azure Dev Tunnel

```
PS C:\Users\s.platas\source\repos\NordicIntegrationSummit\spp-nis25-demo> devtunnel host -p 7053 --allow-anonymous
Hosting port: 7053
Connect via browser: https://q89j12ps-7053.euw.devtunnels.ms
Inspect network activity: https://q89j12ps-7053-inspect.euw.devtunnels.ms

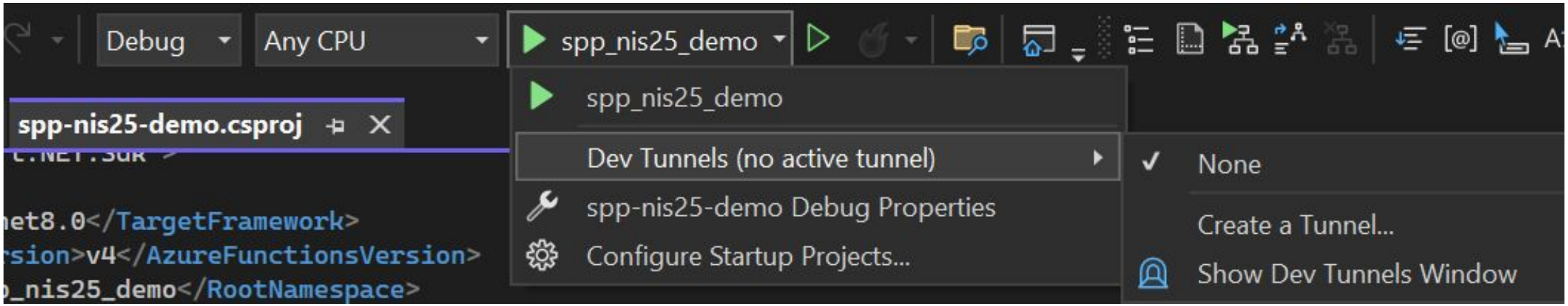
Ready to accept connections for tunnel: amusing-lake-frhvrg4.euw
```



Azure Dev Tunnel

If visual studio 2022+ □ DevTunnels is built in

1. f5
2. debug > dev tunnels > create > public □ public url



ngrok

PREREQUISITES

```
winget install ngrok
ngrok config add-authtoken <YOUR_TOKEN>
```

STEP BY STEP

Func start
- http url – get the port

ngrok http {port}
- result

  Forwarding https://a123b456c.ngrok.io → http://localhost:{port}

Use the public url provided – pautomate, postman…



CLOUDFLARE

PREREQUISITES

```
winget install Cloudflare.cloudflared
```

STEP BY STEP

Func start
- http url - get the port

```
cloudflared tunnel --url http://localhost:7071
```
- result

```
+--------------------------------------------------------------------------------------------+
|  Your quick Tunnel has been created! Visit it at (it may take some time to be reachable):  |
|  https://explore-keyboard-palace-contribute.trycloudflare.com                              |
+--------------------------------------------------------------------------------------------+
```

Kill the process to stop the tunnel.


BEST
PRACTICES

Best practices

- Use https and ephemeral tunnels only
- Never expose production secrets or connections
- Optionally secure with ApiKey or header validation
- Don’t rely on tunnels for long-running integration tests

Takeaways

- Test external integrations instantly
- No redeploy, no cloud delay, full debugger - cut down deployment cycles
- Debug locally with real data
- Tunnels != production – use responsibly
- Use Azure Dev Tunnels for native Microsoft integration
- Use ngrok for visibility & replay
- Remember: tunnels are temporary — secure them and close after use

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Thanks!

Please scan the QR code to rate this session. It’s anonymous, takes 30 seconds, and helps us grow together.

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